

BERENS RIVER, MB, Canada

WMO: 711580

Lat: 52.3597N

Lon: 97.0219W

Elev: 728

StdP: 14.31

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.3	-25.2	-37.4	0.7	-30.1	-32.0	0.9	-25.0	21.4	0.3	19.8	-0.9	3.2	270	0.667

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	81.1	67.6	78.4	66.2	75.8	64.9	70.5	78.1	68.5	75.6	66.7	73.4	8.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.8	105.0	75.2	65.8	97.9	72.8	64.0	91.7	71.1	35.0	78.2	33.3	75.7	31.8	73.2	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.1	18.4	16.2	DB	-37.3	87.7	4.5	3.2	-40.5	90.0	-43.1	91.8	-45.7	93.6	-49.0	96.0	
			WB	-37.6	73.9	4.4	2.9	-40.8	76.0	-43.3	77.7	-45.8	79.3	-49.0	81.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.5	-1.2	2.3	15.8	32.6	46.4	59.2	64.3	62.1	52.7	38.9	21.7	4.9
	DBStd	25.70	14.89	14.33	15.06	12.16	9.47	7.37	5.94	6.28	8.10	8.07	11.85	14.32
	HDD50	7376	1586	1335	1062	528	185	12	0	1	64	355	849	1399
	HDD65	11648	2051	1755	1527	972	580	199	86	132	375	808	1299	1864
	CDD50	1346	0	0	1	6	73	288	445	376	145	12	0	0
	CDD65	141	0	0	0	0	2	26	65	42	6	0	0	0
	CDH74	977	0	0	0	0	30	226	431	255	33	2	0	0
	CDH80	147	0	0	0	0	3	44	64	33	3	0	0	0
	WSAvg	7.5	7.2	6.9	7.5	7.6	7.8	7.4	7.5	7.5	7.7	8.0	7.5	7.2
	PrecAvg	18.4	0.6	0.4	0.8	0.7	1.7	2.6	2.2	2.4	2.6	1.5	1.1	0.7
(15) Precipitation	PrecMax	23.8	2.4	0.8	4.1	1.7	3.5	5.9	5.3	6.1	5.5	5.3	3.7	1.9
	PrecMin	6.9	0.0	0.1	0.0	0.0	0.0	0.7	0.0	0.0	0.4	0.0	0.0	0.0
	PrecStd	3.8	0.6	0.2	0.9	0.5	0.9	1.3	1.4	1.6	1.3	1.3	0.8	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.9	37.9	51.7	67.6	78.0	85.1	85.7	83.9	78.3	66.4	51.9	36.5
		MCWB	33.6	35.0	43.0	51.3	59.3	69.2	70.9	70.1	65.6	57.1	47.7	33.3
	2%	DB	29.6	33.8	44.1	60.7	72.2	79.6	81.2	79.9	73.3	59.2	44.2	31.6
		MCWB	28.1	32.5	37.9	47.5	56.5	65.7	68.4	67.4	63.2	52.1	40.2	30.1
	5%	DB	25.0	30.4	39.6	54.9	67.3	76.0	78.6	76.8	69.4	55.0	40.0	28.7
		MCWB	24.0	28.8	35.3	44.2	53.9	63.5	67.0	65.9	61.0	49.9	37.2	27.8
	10%	DB	20.8	24.4	35.7	49.8	62.4	72.6	75.9	74.0	66.0	51.6	36.8	25.5
		MCWB	20.0	22.9	32.8	41.0	51.9	61.7	66.0	64.8	59.2	47.7	34.8	24.6
	0.4%	WB	33.7	35.7	44.4	52.3	62.2	72.9	74.0	71.9	67.8	58.0	48.3	33.3
		MCDB	34.7	36.8	51.7	65.6	72.3	83.6	82.0	80.3	75.0	65.6	51.4	35.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	28.3	32.8	38.7	47.9	58.8	68.2	71.1	69.7	64.7	53.5	41.0	30.3
		MCDB	29.4	33.9	43.4	58.0	67.4	75.9	78.1	77.4	71.0	58.1	43.5	31.5
	5%	WB	24.1	28.8	35.6	44.9	55.8	65.5	69.0	67.9	62.1	50.3	37.4	27.8
		MCDB	25.0	30.5	39.2	53.7	64.8	72.7	76.0	74.5	67.7	54.5	39.7	28.8
	10%	WB	20.1	23.0	33.1	41.5	53.0	63.4	67.2	66.1	59.8	47.5	34.7	24.7
		MCDB	20.8	24.3	35.6	48.6	61.1	70.7	74.0	72.1	65.7	51.4	36.7	25.6
(35) Mean Daily Temperature Range	5% DB	MDBR	16.4	19.4	20.7	20.5	20.3	19.5	18.6	18.3	17.1	13.2	12.7	15.1
		MCDBR	15.7	16.9	19.9	26.7	27.4	24.2	21.5	21.8	21.4	19.2	14.4	14.0
		MCWBR	15.0	15.1	14.7	16.1	15.2	13.3	12.1	12.3	13.1	13.1	11.6	13.1
	5% WB	MCDBR	16.0	15.6	17.7	25.3	22.6	20.1	19.1	18.9	18.6	17.2	13.8	14.3
		MCWBR	15.3	14.1	13.4	15.9	13.6	11.9	11.7	11.6	12.7	12.4	11.5	13.4
(40) Clear-Sky Solar Irradiance	taub		0.261	0.259	0.305	0.345	0.354	0.374	0.387	0.364	0.332	0.296	0.261	0.238
	taud		2.221	2.279	2.279	2.366	2.422	2.384	2.370	2.461	2.522	2.576	2.417	2.295
	Ebn at Noon		234	276	281	283	285	279	272	274	272	263	239	227
	Edh at Noon		21	26	33	35	34	36	36	31	26	20	17	16
(44) All-Sky Solar Radiation	RadAvg		273	579	977	1469	1735	1816	1809	1504	1045	570	289	206
	RadStd		29	48	98	129	128	130	95	105	115	63	31	29

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.81	-0.97	N/A	N/A	N/A	N/A

Nomenclature: See separate page